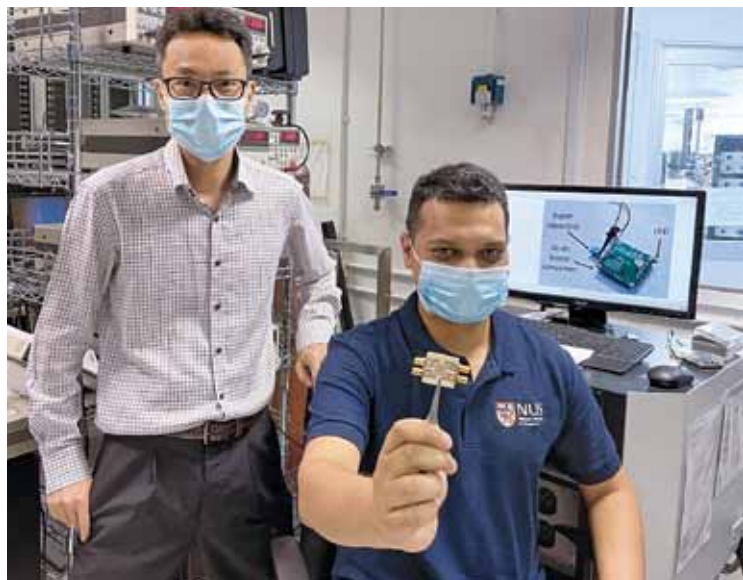


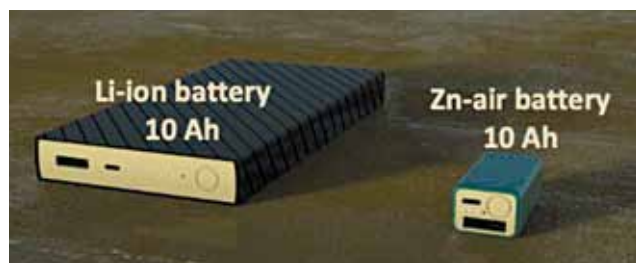
Wi-Fi signals to power small electronics

There has been an increase in the number of Wi-Fi sources that transmit information wirelessly between devices, resulting in the widespread use of the 2.4GHz radio frequency that Wi-Fi uses, with excess signals available to be tapped for alternative uses. To harness this under-utilised source of energy, a research team from the National University of Singapore and Japan's Tohoku University has developed a technology that uses tiny smart devices known as spin-torque oscillators (STOs) to harvest and convert wireless radio frequencies into energy to power small electronics. The research team came up with an array in which eight STOs are connected in series. Using this array, the 2.4GHz radio waves were converted into a direct voltage signal, which was then transmitted to a capacitor to light up a 1.6-volt LED. When the capacitor was charged for five seconds, it was able to light up the same LED for one minute after the wireless power was switched off.

The research breakthrough was achieved by a team led by Prof. Yang Hyunsoo (left). Dr Raghav Sharma (right), the first author of the paper, who is holding a chip embedded with about 50 spin-torque oscillators (Credit: <https://news.nus.edu.sg>)



Ultra-high energy zinc-air pouch battery cells

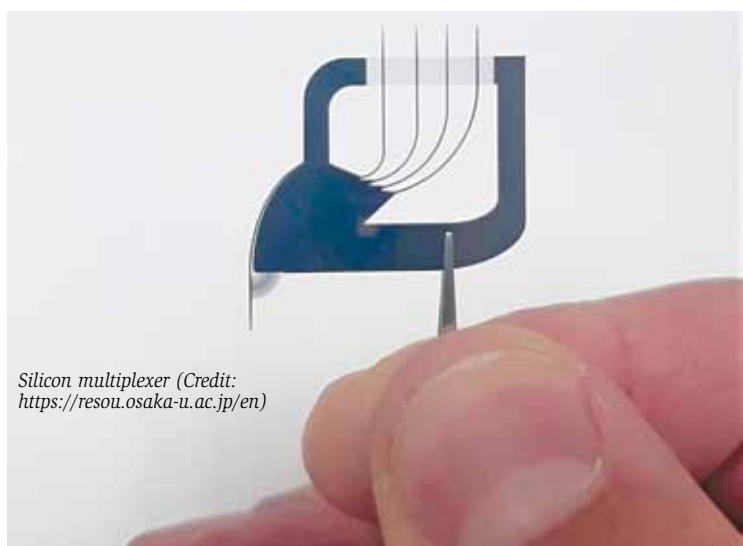


Schematic comparison of energy density for Li-ion battery and all solid-state Zn-air battery (Credit: <https://techxplore.com>)

Researchers at Hanyang University in South Korea have recently designed a new type of zinc-air pouch cell that can outperform other commercially available battery technologies. These pouch cells use (101)-facet copper phosphosulfide [CPS(101)] as a cathode, anti-freezing chitosan-biocellulosics as superionic conductor electrolytes, and patterned zinc as the anode. The zinc-air pouch cells could be commercially viable as they exhibit ultrahigh cell-level energy densities at a wide range of temperatures (-20°C to 80°C), with a high rate capacity of 5-200mA, with a potential for powering drones, electric vehicles, or electric short-distance aircrafts.

Multiplexer using optical tunneling for ultra-high-speed transmission

Compact and practical multiplexers for the terahertz range had not been developed earlier. Now researchers from Osaka University, Japan and the University of Adelaide, Australia have together produced a new multiplexer made from pure silicon for terahertz-range communications in the 300GHz band thanks to a novel optical tunnelling process. The new multiplexer covers a spectral bandwidth that is over 30 times the total spectrum allotted for 4G/LTE and 5G combined. As bandwidth is related to data rate, ultra-high-speed digital transmission is possible with the new four-channel multiplexer. It can potentially support aggregate data rate of 48Gbit/s, equivalent to that of uncompressed 8K ultrahigh definition video being streamed in real time.



Silicon multiplexer (Credit: <https://resou.osaka-u.ac.jp/en>)